

TIC 412057714

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 4.487569 d
Epoch (t0) 3240.0188 BTJD
Duration 0.0840 d
Depth 1.211 % (12,105 ppm)
SNR 189.2

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

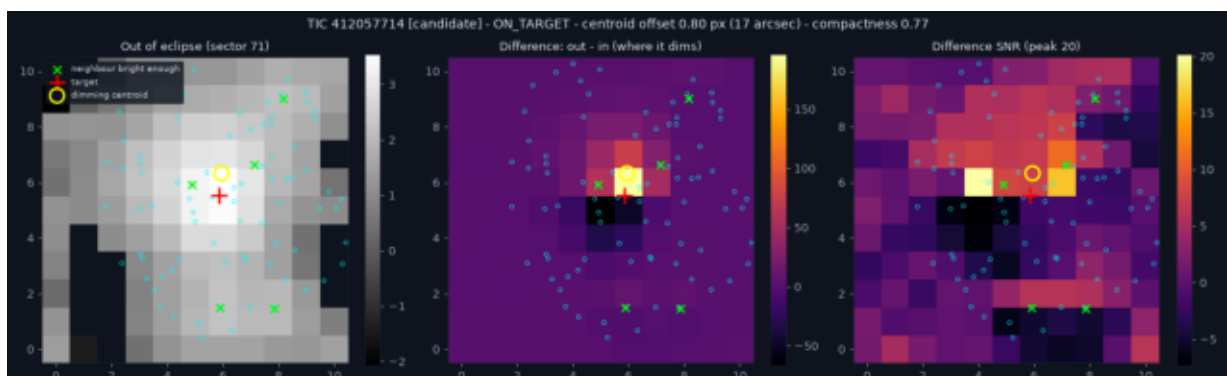
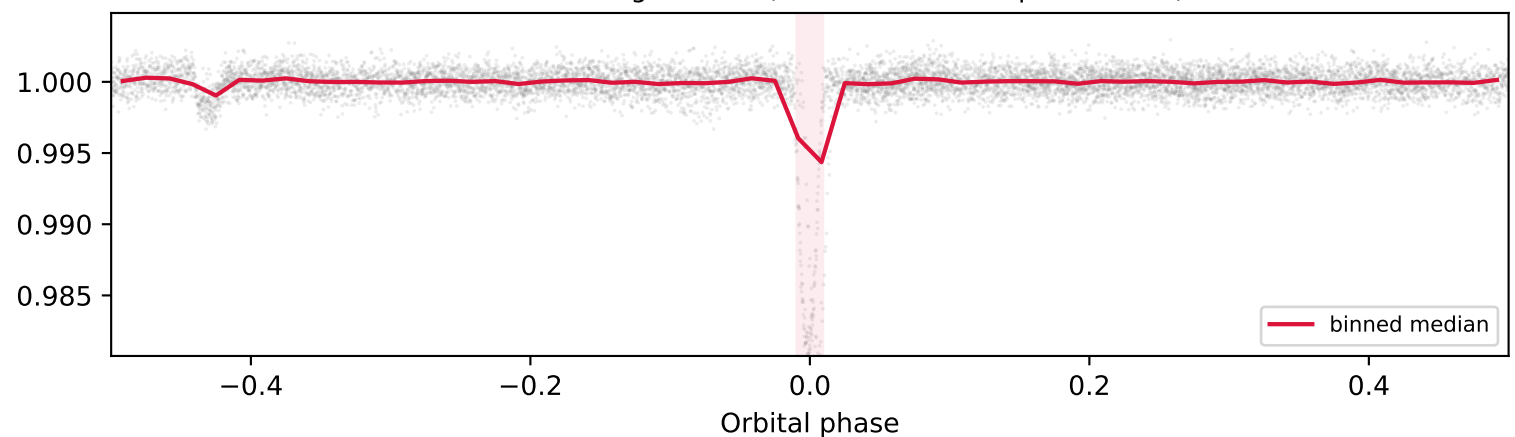
Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.16 (gate ≥ 0.50)
Robust depth sigma 165.8 (gate ≥ 7.0)
Odd/even z-score 2.51 (consistent)
Secondary eclipse detected, phase -0.43, 6.0 sigma

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 71 light-curve dip in this sector: 147.0 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 20.15 (gate ≥ 5.0) Compactness 0.77 (gate ≥ 0.50)
Dimming centroid offset 0.80 px (17 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 412057710
1.12 px away, dTmag +4.02 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.